Equations solved in `EEccz4_BH_puncture.py`

vacuum CCZ4 PINN for Brill–Lindquist / Bowen–York black-hole binaries (state of the code on 2026-10-16)

The network maps (t, x, y, z) to the 22 CCZ4 fields $\left(\tilde{\gamma}_{ij},\ \tilde{A}_{ij},\ \chi,\ \hat{K},\ \Theta,\ \hat{\Gamma}^i,\ \alpha,\ \beta^i\right)$ (6+6+1+1+1+3+1+3). There is no matter: the stress–energy sources τ, S_i, S_{ij} are identically zero. Every field obeys a first-order-in-time equation $\partial_t(\text{field}) = \text{RHS}(\text{fields, spatial derivatives})$; the residual of each one is a row of the loss (`r_ccz4_evolution`). The system is the conformal CCZ4 of Bezares, Palenzuela & Bona 2017 (arXiv:1705.01071, Sec. II), i.e. Alic et al. 2012 with $\chi = \gamma^{-1/3}$, the $\lambda_0 \operatorname{tr}\tilde{A}$ term and κ_c damping of the conformal constraints, with the deliberate differences listed in Sec. 3.

1. Fields, network and metric reconstruction

$\chi = (\det\gamma_{ij})^{-1/3}$, $\tilde{\gamma}_{ij} = \chi\gamma_{ij}$, $\hat{K} = K - 2\Theta$, $\tilde{A}_{ij} = \chi(K_{ij} - \frac{1}{3}\gamma_{ij}K)$; Θ is the Z4 energy-constraint field and $\hat{\Gamma}^i$ the evolved conformal connection, to be compared with the algebraic $\tilde{\Gamma}^i = \tilde{\gamma}^{jk}\tilde{\Gamma}^i_{jk}$. The momentum-constraint violation is

$$Z^i = \frac{\chi}{2}\left(\hat{\Gamma}^i - \tilde{\Gamma}^i\right). \tag{1}$$

Network. A tanh MLP (default 6 hidden layers of 40) with inputs $\big(2(t - t_s)/(t_e - t_s) - 1,\ x/R,\ y/R,\ z/R\big)$, $[t_s, t_e]$ the current window and R=`--R_domain` (24). The output layer starts small (std 10^{-3}) and is shifted to flat space: $\tilde{\gamma}_{ij} \to \delta_{ij}$, $\alpha \to 1$, all others $\to 0$; $\chi \to 1$ only under `--chi_bg_mode physical` (see Sec. 4).

4-metric. For plots only, `adm_to_g4` rebuilds $\gamma_{ij} = \tilde{\gamma}_{ij}/\chi$, $\beta_i = \gamma_{ij}\beta^j$, $g^{tt} = -1/\alpha^2$, $g^{ti} = \beta^i/\alpha^2$, $g^{ij} = \gamma^{ij} - \beta^i\beta^j/\alpha^2$, $\sqrt{-g} = \alpha\sqrt{\gamma}$.

2. Building blocks

2a. Conformal Ricci split

With $\tilde{\Gamma}_{ijk}$ the conformal Christoffel symbol of the first kind (symmetric in its last two indices) and $\tilde{\Gamma}^k_{ij} = \tilde{\gamma}^{kl}\tilde{\Gamma}_{lij}$,

$$\hat{R}^\chi_{ij} = -\tfrac{1}{2}\tilde{\gamma}^{mn}\partial_m\partial_n\tilde{\gamma}_{ij} + \tfrac{1}{2}\Big(\tilde{\gamma}_{ki}\partial_j\hat{\Gamma}^k + \tilde{\gamma}_{kj}\partial_i\hat{\Gamma}^k\Big) + \tfrac{1}{2}\Big(\hat{\Gamma}^k\tilde{\Gamma}_{ijk} + \hat{\Gamma}^k\tilde{\Gamma}_{jik}\Big) + \tilde{\gamma}^{mn}\Big(\tilde{\Gamma}^k_{mi}\tilde{\Gamma}_{jkn} + \tilde{\Gamma}^k_{mj}\tilde{\Gamma}_{ikn} + \tilde{\Gamma}^k_{mi}\tilde{\Gamma}_{knj}\Big), \tag{2}$$

$$R^\chi_{ij} = \frac{1}{\chi}\Bigg[\tfrac{1}{2}\partial_i\partial_j\chi - \tfrac{1}{2}\tilde{\Gamma}^k_{ij}\partial_k\chi - \frac{1}{4\chi}\partial_i\chi\,\partial_j\chi + \frac{1}{\chi}\Big(\tilde{\gamma}_{ik}Z^k\partial_j\chi + \tilde{\gamma}_{jk}Z^k\partial_i\chi\Big) + \tfrac{1}{2}\tilde{\gamma}_{ij}\left(\tilde{\gamma}^{km}\partial_k\partial_m\chi - \frac{3}{2\chi}\tilde{\gamma}^{km}\partial_k\chi\,\partial_m\chi - \hat{\Gamma}^k\partial_k\chi\right)\Bigg]. \tag{3}$$

Because the evolved $\hat{\Gamma}^k$ (not $\tilde{\Gamma}^k$) is used and R^χ_{ij} carries the explicit Z term,

$$\hat{R}^\chi_{ij} + R^\chi_{ij} = R_{ij} + D_iZ_j + D_jZ_i, \qquad \gamma^{ij}\big(\hat{R}^\chi_{ij} + R^\chi_{ij}\big) = R + 2D_iZ^i \tag{4}$$

(arXiv:1705.01071, below their Eq. (syseq2)). The combinations $R_{ij} + 2D_{(i}Z_{j)}$ and $R + 2D_iZ^i$ needed in Sec. 3 are therefore computed directly as Eq. (4); **no separate D_iZ_j or D_iZ^i is added.** (Until 2026-10-10 the code added them again, i.e. it evolved $3D_{(i}Z_{j)}$ in $\tilde{A}_{ij}$ and $R + 4D_iZ^i$ in Θ. Exact solutions have $Z = 0$ and were unaffected; constraint-violating states were not. All runs before that date used the double-counted terms.)

2b. Lapse Hessian and Laplacian

$$D_iD_j\alpha = \partial_i\partial_j\alpha - \Gamma^k_{ij}\partial_k\alpha + \frac{1}{2\chi}\Big(\partial_i\chi\,\partial_j\alpha + \partial_j\chi\,\partial_i\alpha - \tilde{\gamma}_{ij}\,\tilde{\gamma}^{kl}\partial_k\chi\,\partial_l\alpha\Big), \tag{5}$$

$$D^iD_i\alpha = \chi\,\tilde{\gamma}^{ij}\partial_i\partial_j\alpha - \chi\,\tilde{\Gamma}^i\partial_i\alpha - \tfrac{1}{2}\,\tilde{\gamma}^{ij}\partial_i\alpha\,\partial_j\chi. \tag{6}$$

2c. Divisions by χ and α

Every division by χ in the residual (the R^χ terms, $-\frac{3}{2\chi}\tilde{A}^{ij}\partial_j\chi$ and $-\frac{1}{\chi}Z^i(\dots)$ in Eq. (12), and Eq. (5)) uses $\max(\chi, \chi_{\text{floor}})$, `--chi_floor` $= 10^{-5}$ (0 restores the old sign-blind $\sqrt{\chi^2 + 10^{-28}}$). Collocation points are kept at least `--min_puncture_dist` $= 0.02$ from each tracked puncture, where $\chi_{\text{BL}} \approx 2.96 \times 10^{-5}$ at $t = 0$; the script prints this margin at startup. The residual never divides by α: the term $2\alpha\,[-\tilde{\gamma}^{ij}(\Theta/\alpha)\partial_j\alpha]$ of Eq. (12) is written as $-2\,\tilde{\gamma}^{ij}\Theta\,\partial_j\alpha$, which stays correct where an under-trained network has $\alpha < 0$.

3. Evolution equations (vacuum)

With $\mathcal{L}_\beta$ the Lie derivative along β^i, $K \equiv \hat{K} + 2\Theta$, $\mathrm{tr}\tilde{A} = \tilde{\gamma}^{ij}\tilde{A}_{ij}$, $\tilde{A}^{ij}$ raised with $\tilde{\gamma}$, and TF[·] the trace-free part with respect to $\tilde{\gamma}_{ij}$:

$$\partial_t \tilde{\gamma}_{ij} = \mathcal{L}_\beta \tilde{\gamma}_{ij} - \tfrac{2}{3}\tilde{\gamma}_{ij}\partial_k\beta^k - 2\alpha\Big(\tilde{A}_{ij} - \tfrac{1}{3}\tilde{\gamma}_{ij}\mathrm{tr}\tilde{A}\Big) - \tfrac{\alpha}{3}\kappa_c\,\tilde{\gamma}_{ij}\ln(\det\tilde{\gamma}), \tag{7}$$

$$\partial_t \chi = \beta^k\partial_k\chi + \tfrac{2}{3}\chi\Big(\alpha K - \partial_k\beta^k\Big), \tag{8}$$

$$\partial_t \hat{K} = \beta^k\partial_k\hat{K} - D^iD_i\alpha + \alpha\Big[\tfrac{1}{3}K^2 + \tilde{A}_{ij}\tilde{A}^{ij} + \kappa_z(1-\kappa_2)\Theta\Big] + 2Z^i\partial_i\alpha, \tag{9}$$

$$\partial_t \Theta = \beta^k\partial_k\Theta + \tfrac{1}{2}\alpha\Big[R + 2D_iZ^i + \tfrac{2}{3}\hat{K}^2 + \tfrac{2}{3}\Theta(\hat{K} - 2\Theta) - \tilde{A}_{ij}\tilde{A}^{ij}\Big] - Z^i\partial_i\alpha - \alpha\,\kappa_z(2+\kappa_2)\Theta, \tag{10}$$

$$\begin{aligned}\partial_t \tilde{A}_{ij} = {}& \mathcal{L}_\beta\tilde{A}_{ij} - \tfrac{2}{3}\tilde{A}_{ij}\partial_k\beta^k - \tfrac{\alpha}{3}\kappa_c\,\tilde{\gamma}_{ij}\mathrm{tr}\tilde{A} + \chi\,\mathrm{TF}\Big[\alpha\big(R_{ij} + 2D_{(i}Z_{j)}\big) - D_iD_j\alpha\Big] \\ & + \alpha\big(\hat{K}\tilde{A}_{ij} - 2\tilde{A}_{ik}\tilde{\gamma}^{kl}\tilde{A}_{lj}\big),\end{aligned} \tag{11}$$

$$\begin{aligned}\partial_t \hat{\Gamma}^i = {}& \beta^j\partial_j\hat{\Gamma}^i - \hat{\Gamma}^j\partial_j\beta^i + \tfrac{2}{3}\hat{\Gamma}^i\partial_k\beta^k + \tilde{\gamma}^{jk}\partial_j\partial_k\beta^i + \tfrac{1}{3}\tilde{\gamma}^{ij}\partial_j\partial_k\beta^k - 2\tilde{A}^{ij}\partial_j\alpha \\ & + 2\alpha\Big[\tilde{\Gamma}^i_{jk}\tilde{A}^{jk} - \tfrac{3}{2\chi}\tilde{A}^{ij}\partial_j\chi - \tfrac{2}{3}\tilde{\gamma}^{ij}\partial_j\hat{K} - \tfrac{1}{3}\tilde{\gamma}^{ij}\partial_j\Theta - \tfrac{1}{\chi}Z^i\big(\kappa_z + \tfrac{2}{3}K\big)\Big] - 2\,\tilde{\gamma}^{ij}\Theta\,\partial_j\alpha,\end{aligned} \tag{12}$$

$$\partial_t\alpha = \beta^i\partial_i\alpha - 2\alpha f_\alpha\hat{K} \qquad \text{(1+log slicing)}, \tag{13}$$

$$\partial_t\beta^i = \beta^j\partial_j\beta^i + \tfrac{3}{4}f_\beta\hat{\Gamma}^i - \eta\,\beta^i \qquad \text{(integrated Gamma-driver)}. \tag{14}$$

In Eqs. (10) and (11), $R + 2D_iZ^i$ and $R_{ij} + 2D_{(i}Z_{j)}$ are evaluated as in Eq. (4). The 18 rows (7)–(12) form the "Einstein" loss and the 4 rows (13)–(14) the "gauge" loss.

Parameters and defaults. $\kappa_z, \kappa_c, \kappa_2$ (`--kappa_z`/`--kappa_c`/`--kappa_2`, all 0: no constraint damping); $f_\alpha = f_\beta = 1$, $\eta = 0.1$ (`--f_alpha`/`--f_beta`/`--eta`); `--gauge static` freezes $\alpha = 1$, $\beta^i = 0$ (debugging).

Differences from arXiv:1705.01071 (deliberate). (i) Lapse: $-2\alpha f_\alpha\hat{K}$ equals the paper's $-\alpha^2 f\hat{K}$ with $f = 2/\alpha$. (ii) Shift: integrated Gamma-driver, no B^i field and no β_0 term. (iii) $\hat{\Gamma}^i$, not $\tilde{\Gamma}^i$, in the shift terms of Eq. (12) (Alic's $\kappa_3 = 1$); κ_2 is an addition. (iv) Boundary speed $c = 1$ for all fields (Sec. 6).

Verification (`handoff_EEccz4_BH_puncture/tests/`). All 22 rows match an independent implementation in Alic et al.'s variables ($\mathrm{tr}K, \Theta$; physical Ricci, $D_iD_j\alpha$, D_iZ_j by autodiff of $\tilde{\gamma}_{ij}/\chi$) to $\sim 10^{-16}$ on random fields with $Z, \Theta, \beta \neq 0$ and nonzero $\kappa_z, \kappa_c, \kappa_2, f_\alpha, f_\beta, \eta$; static isotropic Schwarzschild gives 0 in all 22 rows and Kerr ($a = 0.6$, Kerr–Schild) 0 in the 18 Einstein rows; flat space gives exactly 0.

4. Moving χ background (`--chi_bg_mode moving`, default)

The network's χ channel is the smooth deviation from an analytic Brill–Lindquist background that follows the tracked punctures:

$$\chi(t,\vec{x}) = \chi_{\mathrm{net}}(t,\vec{x}) + \psi_{\mathrm{bg}}^{-4}, \qquad \psi_{\mathrm{bg}} = 1 + \sum_A \frac{m_A}{2\,|\vec{x} - \vec{x}_A(t)|}, \tag{15}$$

with $\vec{x}_A(t) = (x_A(t), y_A(t), 0)$ interpolated from the shift-driven trajectory of Sec. 8 and held fixed with respect to the optimizer (stop-gradient). It is refreshed at every resample with `--moving_bg_refresh_steps` RK4 steps. This χ is the one used in the residual, the boundary condition and the diagnostics. `--chi_bg_mode physical`: the network outputs χ itself.

5. Initial data (Brill–Lindquist / Bowen–York)

Conformally flat multi-puncture data, punctures on the x-axis at $\mp$`--star_separation`$/2$ (default 4, `--n_stars` 2):

$$\psi = 1 + \sum_A \frac{m_A}{2r_A}, \qquad \gamma_{ij} = \psi^4 \delta_{ij}, \qquad \alpha = 1/\psi, \qquad \beta^i = 0, \tag{16}$$

masses `--BL_masses` (0.5,0.5). Each puncture may carry an in-plane momentum $P_A = (P_A^x, P_A^y, 0)$ (`--BL_momenta`, default 0,0, and `--BL_momenta_y`, default $-0.16, 0.16$: a bound orbit) and a spin S_A (`--BL_spins`, default 0). With $n_A^i = (x - x_A)^i / r_A$ built from a max-floored radius $r_A = \sqrt{\max(r^2, \texttt{r_eps}^2)}$,

$$\bar{A}_A^{ij} = \frac{3}{2r_A^2}\Big[P_A^i n_A^j + P_A^j n_A^i - \big(\delta^{ij} - n_A^i n_A^j\big)(P_A \cdot n_A)\Big] + \frac{3}{r_A^3}\Big[\epsilon^{ikl} S_{A,k} n_{A,l}\, n_A^j + \epsilon^{jkl} S_{A,k} n_{A,l}\, n_A^i\Big], \tag{17}$$

$$K_{ij} = \psi^{-2} \sum_A \bar{A}_{A,ij}. \tag{18}$$

This K_{ij} satisfies the momentum constraint exactly for any ψ and is trace-free; the Hamiltonian constraint is not solved (no u correction), so its residual is left to the loss (it vanishes only for $P_A = S_A = 0$). The CCZ4 data follow algebraically: $\chi = \psi^{-4}$, $\tilde{\gamma}_{ij} = \delta_{ij}$, $\tilde{A}_{ij} = \chi(K_{ij} - \frac{1}{3}\gamma_{ij}K)$, $\hat{K} = K = 0$, $\Theta = 0$, $\hat{\Gamma}^i = \tilde{\Gamma}^i$ (so $Z^i = 0$), $\alpha = 1/\psi$, $\beta^i = 0$ (`compute_ccz4_ic`).

Windows. Time is split into `--N_window` windows (200 of width 0.25 for `--tfinal` 50). Window 0 starts from the data above; every later window starts from the previous window's network at its end time (`--reinit_mode warm`: the weights are inherited too). `--ic_mode soft` (default) imposes this by the loss term of Sec. 7; `--ic_mode hard` uses $f = f_{\rm IC} + \tau\,\mathrm{NN}$, $\tau = (t - t_s)/(t_e - t_s)$.

6. Boundary conditions

Sommerfeld (`--metric_bc sommerfeld`, default) for all 22 fields X at $r = R$, towards the flat values X_∞ ($\tilde{\gamma}_{ij} \to \delta_{ij}$, $\chi, \alpha \to 1$, all others $\to 0$):

$$\partial_t X + c\,\frac{x^i}{r}\partial_i X + c\,\frac{X - X_\infty}{r} = 0, \qquad c = 1. \tag{19}$$

The 1+log gauge speed $\sqrt{2\alpha}$ is ignored. `--metric_bc dirichlet` fixes every field to its $t = 0$ value at the boundary.

7. Loss

With $r^{(k)}$ the residual of Eq. k of Sec. 3, interior points $\mathcal{I}$, the window-end surface $\mathcal{T}$ ("tapa"), boundary points $\mathcal{B}$ and window-start points $\mathcal{S}$:

$$\begin{aligned} L = {} & \lambda_E\Big[\big\langle w_c\,|r_{1..18}|^2\big\rangle_{\mathcal{I}} + \lambda_{\text{tapa}}\,w_T\big\langle|r_{1..18}|^2\big\rangle_{\mathcal{T}}\Big] + \lambda_G\Big[\big\langle w_c\,|r_{19..22}|^2\big\rangle_{\mathcal{I}} + \lambda_{\text{tapa}}\,w_T\big\langle|r_{19..22}|^2\big\rangle_{\mathcal{T}}\Big] \\ & + \lambda_{bc}\big\langle|r_{BC}|^2\big\rangle_{\mathcal{B}} + \lambda_{g0}\Big[\big\langle|u-u_{\text{IC}}|^2\big\rangle_{\mathcal{S}} + \lambda_{\log}\big\langle(\ln\chi-\ln\chi_{\text{IC}})^2+(\ln\alpha-\ln\alpha_{\text{IC}})^2\big\rangle_{\mathcal{S}}\Big] \\ & + w_{\text{pos}}\big\langle\text{relu}(-\chi)^2+\text{relu}(-\alpha)^2\big\rangle_{\mathcal{I}}, \end{aligned} \tag{20}$$

$\langle\cdot\rangle$ the mean over points and components. Defaults: $\lambda_E = \lambda_G = \lambda_{bc} = 1$, $\lambda_{\text{tapa}} = 2$, $\lambda_{g0} = 10$, $\lambda_{\log} = 0$ (`--lam_ic_log`), $w_{\text{pos}} = 10$. *Causal weights* (`--causal_weight`, on): the window is cut into `--causal_buckets` (8) time buckets and each interior point is weighted by $w_c = \exp(-\epsilon\sum_{b'<b}\bar{r}^2_{b'})$, the stop-gradient sum of the mean squared residual of the earlier buckets (ϵ=`--causal_eps`, 1); the tapa term gets $w_T = \exp(-\epsilon\sum_b \bar{r}^2_b)$.

8. Puncture tracking

`--bh_center_mode shift_driven` (default): each puncture moves with

$$\frac{dx_A}{dt} = -\beta^x(x_A, y_A, \varepsilon, t), \qquad \frac{dy_A}{dt} = -\beta^y(x_A, y_A, \varepsilon, t), \qquad \varepsilon = 0.03, \tag{21}$$

integrated with RK4 (`--track_steps` 1000) through the trained network of each window; the positions are never trainable. They steer the focused sampling and the moving background of Sec. 4. `chi_min` and `lapse_min` relocate the centres from minima of χ or α in the x–y plane instead (not compatible with the moving background); `fixed` keeps them at $t = 0$.

9. Optimizer

Per window: optional Adam (`--EPOCHS_ADAM`, 0) then quasi-Newton restarts of `--N_change` (250) iterations on a fixed batch, with a fresh batch between restarts and the inverse Hessian H carried over (`--EPOCHS_QN` 12000, i.e. 48 restarts). Update (`--qn_update`): self-scaled Broyden `ssbroyden2` (default), `bfgs`, or SCORE (`score`: SSBroyden on the shifted pair $(s,\ y + \mu_k s)$, $\mu_k = \text{clip}(\eta_S\lambda_k, \mu_{\min}, \mu_{\max})$, $\lambda_k = \sqrt{g^\top H g}$, first line-search trial $1/(1+\lambda_k)$; defaults $\eta_S = 0.1$, $\mu_{\min} = 10^{-8}$, $\mu_{\max} = 10^{-4}$). Line search (`--line_search`): strong Wolfe (default, $c_1 = 10^{-4}$, c_2=`--wolfe_c2`=0.9) or Armijo backtracking. `--h0_scaling` 1 (default) rescales an identity H to $(y\cdot s/y\cdot y)\,I$ after the first step (Nocedal & Wright eq. 6.20). Safeguards: a non-finite restart is undone (weights restored, $H = I$, new batch; the window ends after 3 in a row); the excursion guard discards a restart whose loss exceeds `--excursion_ratio` (100) times the previous one, or, for the first restart of a later window, max(`--excursion_abs`, 100× the previous window's final loss), up to `--excursion_retries` (3) times; a stall (no relative improvement >`--min_delta` for `--patience_bfgs` restarts, and best loss ≤`--es_min_loss` if that is > 0) first resets $H = I$ twice and then ends the window. With the defaults (patience 500 > 48 restarts) the stall rule never fires.

10. Diagnostics

Θ and Z^i (Eq. (1)) are the constraint monitors. `hamiltonian_constraint()` returns Θ (the plots call it "H"), not $R+K^2-K_{ij}K^{ij}$. Puncture health is read from α at the punctures (`puncture_track.txt`) and `puncture_strength.csv`.